

Model Results and Conclusion

Project: Plant disease vision models

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Model Architectures Overview

- **Swin-HViT:** A hybrid model combining Swin Transformers (for local, CNN-like shifted-window understanding) with standard Vision Transformers (for global context).
- **ViT / DeiT:** Standard Vision Transformers (and Data-efficient variants) that rely on global self-attention mechanisms.
- **Bottleneck ViT (JiT-style):** A modified ViT that introduces a low-dimensional bottleneck layer right at the patch embedding stage to compress the representation and smooth the learning manifold.
- **VMamba:** A Vision State Space Model (SSM) using 2D cross-scan mechanisms to achieve linear complexity, designed as a faster alternative to attention (though severely hindered by Kaggle CUDA compilation issues in these tests).
- **MLP-Mixer:** An attention-free architecture that relies entirely on Multi-Layer Perceptrons to mix spatial locations and feature channels independently.
- **FocalNet:** Replaces standard self-attention with focal modulation networks, using hierarchical contextualization and gated aggregation for efficient feature extraction.

Comprehensive Experimental Results (Feb-Mar 2026)

Model Architectures Overview

Model Architecture	Description & Key Features
Swin-HViT	A hybrid model combining a Swin Transformer (for local, CNN-like shifted-window understanding) with a standard Vision Transformer (for global context), fused via simple concatenation.
ViT / DeiT	Standard Vision Transformers (and Data-efficient variants) that process images as patches and rely on global self-attention mechanisms.
Bottleneck ViT (JiT)	A modified ViT that introduces a low-dimensional bottleneck layer (e.g., dim 16, 32, 64) at the patch embedding stage to compress representations and smooth the learning manifold.
VMamba	A Vision State Space Model (SSM) using 2D cross-scan mechanisms (unwinding patches in 4 directions) to achieve linear complexity and global context faster than attention.
MLP-Mixer	An attention-free architecture that relies entirely on Multi-Layer Perceptrons to mix spatial locations and feature channels independently, maintaining a constant hidden size across layers.
FocalNet	An architecture replacing standard self-attention with focal modulation networks, utilizing hierarchical contextualization, gated aggregation, and element-wise modulation for efficient feature extraction.

Model Architecture	Experiment Description & Dataset	Peak Accuracy
Swin-HViT	Trained on Corn, Validated on PlantDoc (Zero-shot)	35.0%
Swin-HViT	Trained on PlantVillage, Validated on PlantDoc (Zero-shot)	30.0%
Swin-HViT	Mixed Training (PV + PD), Validated on held-out PD	62.75%
Bottleneck DeiT-Tiny (Dims 16/32/64)	Trained on PV, Validated on PD (Zero-shot)	~11.0%
Bottleneck DeiT-Tiny (Dim 64)	Mixed Training (PV + PD), Validated on held-out PD	23.5%
Swin-HViT	Rice disease dataset (Pre-trained baseline)	89.0%
ViT-Base	Rice disease dataset (Pre-trained)	87.5%
Swin-Tiny	Rice disease dataset (Pre-trained)	88.0%
DeiT-Tiny	Rice disease dataset (Pre-trained)	81.0%
DeiT-Tiny	Rice disease dataset (Trained from scratch, epoch 20)	75.0%
Bottleneck DeiT-Tiny (Dim 64)	Rice disease (Trained from scratch, epoch 60)	82.0%
Bottleneck DeiT-Tiny (Dim 32)	Rice disease (Trained from scratch, epoch 20)	79.0%
Bottleneck DeiT-Tiny (Dim 16)	Rice disease (Trained from scratch, epoch 40)	81.0%
Bottleneck DeiT-Tiny (Dim 8)	Rice disease (Trained from scratch, epoch 20)	73.0%
VMamba	Rice disease (Failed custom CUDA compile, unoptimized speed)	75.0%
MLP-Mixer (Base)	Rice disease (Trained from scratch)	65.5%
MLP-Mixer + AdamP	Rice disease (Trained from scratch, stagnated)	73.5%
MLP-Mixer + SAM	Rice disease (Trained from scratch, balanced F1)	80.0%
FocalNet-Tiny	Rice disease (Trained from scratch)	72.0%

Model Architecture	Experiment Description & Dataset	Peak Accuracy
FocalNet-Tiny	Rice disease (Pre-trained)	88.0%
Swin-HViT	5 difficult Rice classes (Pre-trained, peaked before overfitting)	95.0%
Swin-HViT + SAM	5 difficult Rice classes (Regularized, degraded F1)	72.5%
Swin-HViT	17 Rice classes (Removed highly confused Leaf Smut & Blast)	92.8%
MLP-Mixer + SAM	17 Rice classes (Removed Leaf Smut & Blast)	75.0%
Swin-HViT	Fruit disease dataset (17 classes, Pre-trained)	97.3%
MLP-Mixer + SAM	Fruit disease dataset (17 classes, Trained from scratch)	92.0%
Swin-HViT	Potato Tomato dataset (Pre-trained)	64.5%

Conclusion

Over the past six weeks, experimentation spanned a wide array of architectures, datasets, and regularization techniques to identify the optimal pipeline for plant disease classification. Early tests confirmed that zero-shot transfer across datasets (like PlantVillage to PlantDoc) yields highly degraded results (often dropping to 11-35%), underscoring the necessity of fine-tuning hybrid models on target data.

Architectural exploration yielded significant insights. Implementing a low-dimensional bottleneck in Vision Transformers (JiT-style) improved minority class F1 scores without pre-training, peaking at 82% (Dim 64). State Space Models (VMamba) proved practically incompatible with the Kaggle environment due to CUDA compilation failures. However, exploring alternative mechanisms like MLP-Mixers combined with Sharpness-Aware Minimization (SAM) successfully bypassed local minima and handled class imbalances remarkably well without pre-training, achieving an impressive 92.0% on Fruit diseases. FocalNet also showed strong promise when pre-trained, reaching 88.0%.

Ultimately, **Swin-HViT stands out as the undisputed top performer** across all benchmarks. It achieved the absolute highest accuracy of **97.3%** on the Fruit dataset, reached **95.0%** when isolated to the 5 most difficult Rice disease classes, and secured **92.8%** on the 17-class Rice dataset when highly confused classes were removed. Interestingly, while SAM regularization heavily benefits models trained from scratch like the MLP-Mixer, it actively degraded the performance of Swin-HViT. Future work should focus on utilizing Swin-HViT for primary deployment, isolating highly confused classes (e.g., Leaf Smut and Leaf Blast, which severely drag overall accuracy down) into specialized secondary classifiers to push performance limits even further.